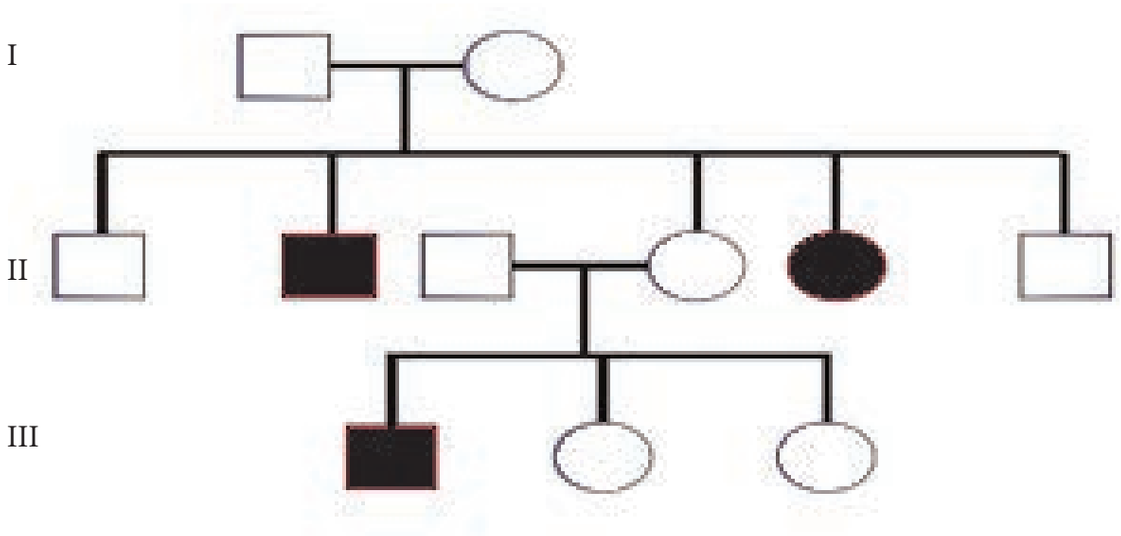
21. AUTOSOMAL DISEASE –SICKLE CELL ANEMIA

IDENTIFICATION

The given pedigree chart is identified as the genetic disease sickle cell anaemia,.

COMMENTS

1. It is an autosome linked recessive trait that can be transmitted from parents to the offspring, if both the parents are carriers for the gene (heterozygous).
2. The genotype Hb^S Hb^S causes sickle cell anemia, while the genotype Hb^A and Hb^S individuals appear apparently unaffected but they are carrier of the disease.
3. Sickle cell anemia is a classical example of point mutation.
4. The defect is caused by the substitution of Glutamic acid by valine at the sixth position of the beta chain of the haemoglobin molecules.



DISCUSSION QUESTIONS

1. Observe the given pedigree chart and give reasons for the occurrence of the disease in the second generation.
2. Will males and females be equally affected in this type of inheritance? Give reasons.
3. What is the genotype of the diseased and carrier sickle sell individual?
4. How is Sickle cell anemia disease caused?

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